

# Malware

## Lesson 42 — The Malware Umbrella

JKSSB Computer Awareness

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# What “Malware” Means

- ▶ **Malware** is short for **malicious software**: any program written to harm a computer, steal from it, or misuse it.
- ▶ The official syllabus names **computer virus and latest anti-virus** explicitly; the exam also tests the neighbours: worm, Trojan, ransomware, spyware, adware, and more.
- ▶ One umbrella word, many behaviours: some **damage** files, some **steal** data, some **demand money**, some **hide** the attacker.
- ▶ The exam rarely asks for stories; it asks **which behaviour belongs to which name**.

## The one idea

Learn each type by its method: how it spreads, what it wants, and what gives it away.

# The Umbrella Map

| Type                             | What it does                         | One-line sign               |
|----------------------------------|--------------------------------------|-----------------------------|
| Virus                            | Attaches to files, needs a click     | Infected attachment runs    |
| Worm                             | Self-spreading over networks         | Travels alone, no host file |
| Trojan                           | Pretends to be useful software       | User installs it willingly  |
| Ransomware                       | Locks files, demands payment         | "Pay to unlock" screen      |
| Spyware / key-logger             | Watches and steals quietly           | Stolen passwords            |
| Adware                           | Floods the screen with ads           | Pop-ups everywhere          |
| Rootkit / back-door              | Hides the attacker, keeps entry open | Nothing visible             |
| Bot / logic bomb / cryptojacking | Misuses the machine                  | Slow PC, timed trigger      |

Details follow; this table is the map to keep in mind.

# Virus

- ▶ A **virus** is malware that **attaches itself to a host** program or file and **needs human action** to run.
- ▶ It spreads when the infected file is opened, shared, or copied: email attachment, pen drive, downloaded file.
- ▶ It can damage, delete, or corrupt files once it runs.
- ▶ Key exam point: a virus **cannot spread by itself**; it needs a host file and a user to launch it.

Keep the pair straight

Virus = needs a host file **and** a human click to spread.

# Worm

- ▶ A **worm** is malware that **spreads by itself over a network**, with no host file and no human click.
- ▶ It copies itself from one machine to another through network connections and security gaps.
- ▶ It mainly harms by **consuming network and system resources**: the network slows down, machines slow down, services jam.
- ▶ Key exam point: a worm is **self-replicating and standalone**; that is what separates it from a virus.

Keep the pair straight

Worm = travels alone over the network; needs **no** host and **no** click.

# Trojan (Trojan Horse)

- ▶ A **Trojan** (Trojan horse) **pretends to be useful software**: a free game, a useful tool, a harmless attachment.
- ▶ The user **installs or opens it willingly**, and the hidden malicious part runs with it.
- ▶ Unlike a virus or worm, a Trojan **does not replicate itself**; it relies on tricking the user once per machine.
- ▶ Once inside, it can steal data, open a **backdoor**, or install further malware such as spyware or ransomware.

## Trap alert

Trojan = disguised as legitimate; it **does not self-replicate**.

## Virus vs Worm vs Trojan

| Point             | Virus       | Worm             | Trojan                  |
|-------------------|-------------|------------------|-------------------------|
| Needs host file   | Yes         | No               | No (pretends to be one) |
| Spreads by itself | No          | Yes              | No                      |
| Needs user action | Yes         | No               | Yes (install/open)      |
| Main harm         | File damage | Network slowdown | Hidden entry/theft      |

### The core exam table

Virus needs host + click; worm needs neither; Trojan tricks the user and does not replicate.

# Ransomware

- ▶ **Ransomware locks** the victim's files by **encryption** or locks the whole screen, then **demands payment** (ransom) for the unlock key.
- ▶ The signature sign is the **“pay to unlock” message** with a countdown or payment address.
- ▶ It typically arrives like a Trojan: a harmful attachment or link the user opens.
- ▶ Defence the exam expects: regular **offline backups**, updated software, and **anti-virus** software — paying does not guarantee recovery.

## Office desktop example

A clerk opens an invoice attachment; files turn unreadable and a message demands payment to restore them — that is ransomware.



# Spyware

- ▶ **Spyware** secretly **watches** what the user does and sends it to someone else: browsing habits, logins, personal details.
- ▶ It **hides quietly** and tries not to damage anything, so the victim notices nothing.
- ▶ It often arrives bundled with free downloads or deceptive installers.
- ▶ Purpose is **theft by observation**: credentials, bank details, and personal data.

Keep the pair straight

Spyware = silent watcher; its goal is stolen information, not visible damage.

# Adware and Keylogger

- ▶ **Adware** floods the machine with **unwanted advertisements**: pop-ups, banners, redirected searches.
- ▶ Its motive is usually money from ads, but heavy adware also **slows** the machine and can carry spyware with it.
- ▶ A **keylogger** records **every key pressed** and sends the log to the attacker — a focused spying tool for passwords and card numbers.
- ▶ Keyloggers may be software, or small **hardware** devices fitted between keyboard and computer.

## Neighbour line

Adware announces itself with ads; spyware and keyloggers hide and steal.

## Spyware vs Adware

| Point             | Spyware            | Adware             |
|-------------------|--------------------|--------------------|
| Visible to user   | No (hides)         | Yes (pop-ups, ads) |
| Main goal         | Steal data quietly | Show ads for money |
| Gives itself away | Stolen accounts    | Flooded screen     |

The second core table

Spyware hides and steals; adware shows itself and advertises.

# Rootkit and Backdoor

- ▶ A **rootkit** is malware that **hides** itself and other malware deep inside the system, often with administrator-level control.
- ▶ Its job is **concealment**: files, processes, and network activity are kept invisible to the user and to ordinary tools.
- ▶ A **backdoor** is a **secret entry path** left into the system, bypassing normal login and checks.
- ▶ Trojans and rootkits commonly **install backdoors** so the attacker can return later without being noticed.

Keep the pair straight

Rootkit = hides the intruder; backdoor = the hidden door the intruder keeps.

# Bot, Botnet, Logic Bomb, Cryptojacking

- ▶ A **bot** is a compromised machine controlled remotely; a **botnet** is a **network of such machines** (robot + network) obeying one master, used for spam and large-scale attacks.
- ▶ A **logic bomb** is malicious code that **waits for a trigger**: a date, an event, or an action — then strikes.
- ▶ **Cryptojacking** secretly uses the victim's computer to **mine cryptocurrency**, so the machine runs hot and slow while the attacker earns.
- ▶ Common thread: the machine is **misused** while the owner notices only slowness or a timed surprise.

# How Malware Reaches You — and How to Stop It

- ▶ Common entry routes: harmful **email attachments and links**, infected pen drives, free-software bundles, and unpatched systems.
- ▶ **Anti-virus** software detects, blocks, and removes known malware; keep it and the operating system **updated**.
- ▶ Practical habits: do not open unknown attachments, scan pen drives, download from trusted sources, and keep **offline backups**.
- ▶ A **firewall** filters network traffic; it is a barrier, not a cleaner — it does not remove malware already inside.

## Exam line

Anti-virus removes malware; firewall filters traffic; backups defeat ransomware.

## Trap Alert: Malware Facts to Keep Straight

- ▶ **Virus** needs a host + click; **worm** needs neither; **Trojan** disguises itself and does not replicate.
- ▶ **Spyware** hides and steals; **adware** shows ads.
- ▶ **Ransomware** encrypts and demands payment; **keylogger** records keystrokes.
- ▶ **Rootkit** hides; **backdoor** is the secret re-entry; **botnet** is a network of controlled machines.
- ▶ **Logic bomb** waits for a trigger; **cryptojacking** mines cryptocurrency on your machine.

## Lesson 42: Recall

- ▶ Malware = malicious software; the umbrella covers virus, worm, Trojan, ransomware, spyware, adware, rootkit, keylogger, bot/botnet, backdoor, logic bomb, and cryptojacking.
- ▶ Virus vs worm vs Trojan: host + click, neither, disguise without replication.
- ▶ Spyware vs adware: silent theft vs visible ads; keylogger is a focused spying tool.
- ▶ Rootkit hides the intruder; backdoor keeps the door open; botnet is a controlled network; logic bomb waits for its trigger; cryptojacking mines on your machine.
- ▶ Protection: updated anti-virus, firewall, careful clicks, trusted downloads, offline backups.